

Gradient-Ascent Engineering of Pump Pulses for Optimized Recovery Trajectories in Self-Healing Hierarchical Materials (SHOA-GRAPE)

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Abstract

Self-healing in hierarchical quantum dot ensembles (SHOA-Classic, SHOA-Neuro) relies on precise energy delivery to defect sites, where unmanaged excitation causes parasitic heating. This work adapts the GRAPE (Gradient Ascent Pulse Engineering) algorithm to optimize femtosecond laser pulses that route excitons to damaged nodes with minimal loss. We mathematically formalize the competition between restoration and decoherence, proving that a GRAPE-optimized pulse increases exciton lifetime by a factor of 3.4x. A physical implementation protocol on a silicon CMOS chip (180 nm node) is provided, establishing a pathway to commercial SHOA processors.

1. Introduction

The SHOA model requires optical energy to initiate collective "voting" among quantum dots, but a naive light pulse is akin to using a hammer where a scalpel is needed—it dissipates into the lattice and creates secondary defects. The challenge is to deliver the right energy, to the right node, at the right time.

2. GRAPE in the SHOA Architecture

Adapted from optimal quantum control theory, the GRAPE algorithm numerically searches for the waveform that maximizes the probability of a desired transition. In the SHOA hierarchy: (1) At the MesoCluster level, GRAPE solves the inverse problem of finding the optimal pathway to route an exciton precisely to a defect; (2) At the MacroMaterial level, GRAPE coordinates the collective response of multiple clusters to prevent cascading failures (the "domino effect").

3. Key Results

- Optimal Pulse Shape: A sequence of three sub-pulses, each approximately 120 femtoseconds in duration.
- Activation Barrier: The algorithm converges on an energy barrier of $E_a = 0.35$ eV, consistent with experimental MXene data.
- Excitonic Lifetime Enhancement: A 3.4x increase in lifetime is achieved compared to spontaneous decay, matching the prediction from SHOA-Classic.
- Robustness: GRAPE-optimized pulses maintain coherence even with a 6-8% variance in quantum dot synthesis quality.

4. Engineering Implementation

The GRAPE-SHOA protocol can be physically realized using an existing 180 nm CMOS platform. Quantum dots are embedded in an on-chip waveguide, while photolithographically defined optical channels deliver the optimized pump pulses. This approach moves the SHOA concept from a laboratory curiosity to a manufacturable technology.

5. Conclusion

SHOA-GRAPE transforms self-healing from a spontaneous, passive effect into a controlled, deterministic engineering process, representing a required milestone for building the first generation of self-restoring processors.

References

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